

LECTURE SHEET

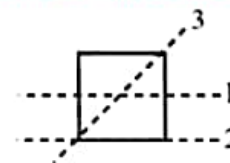
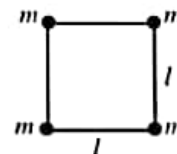
EXERCISE-I

(Moment of Inertia)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Figure shows a square with point masses 'm' at corners. M.I. of the system about x-axis passing through the centre, and M.I. of system about y-axis passing through centre is
 1) ml^2, ml^2 2) $\frac{ml^2}{2}, \frac{ml^2}{2}$ 3) $\frac{ml^2}{4}, \frac{ml^2}{4}$ 4) $\frac{ml^2}{16}, \frac{ml^2}{16}$
- In the above problem M.I. of system about z-axis passing through centre is ____
 1) $2ml^2$ 2) $\frac{ml^2}{2}$ 3) $\frac{ml^2}{4}$ 4) $\frac{ml^2}{16}$
- A thin rod of mass $6m$ and length $6L$ is bent into regular hexagon. The M.I. of the hexagon about a normal axis to its plane and through centre of system is
 1) mL^2 2) $3mL^2$ 3) $5mL^2$ 4) $11mL^2$
- Two uniform thin rods each of mass 'm' and length 'L' are arranged to form a cross. The moment of inertia of the system about an angular bisector is
 1) $\frac{mL^2}{12}$ 2) $\frac{mL^2}{6}$ 3) $\frac{mL^2}{3}$ 4) $\frac{2mL^2}{3}$
- Four rods of equal length l and mass m each form a square as shown in figure. The moment of inertia of the system about the axis 1 axis 2 and axis 3 is
 1) $\frac{2}{3}ml^2, \frac{5}{3}ml^2, \frac{2}{3}ml^2$ 2) $\frac{5}{3}ml^2, \frac{10}{3}ml^2, \frac{5}{3}ml^2$
 3) $\frac{1}{3}ml^2, \frac{5}{3}ml^2, \frac{1}{3}ml^2$ 4) $\frac{10}{3}ml^2, \frac{2}{3}ml^2, \frac{10}{3}ml^2$



LEVEL-II (ADVANCED)

Straight Objective Type Questions

- A quarter sector is cut from a uniform circular disc of radius R . This sector has mass M . It is made to rotate about a line perpendicular to its plane and passing through the centre of the original disc. Its moment of inertia about the axis of rotation is
 a) $MR^2/2$ b) $MR^2/4$ c) $MR^2/8$ d) $\sqrt{2} MR^2$
- A disc of mass 'm' and radius R has a concentric hole of radius 'r'. Its M.I. about an axis through centre and normal to its plane is
 a) $m/2 (R-r)^2$ b) $m/2 (R^2-r^2)$ c) $m/2(R+r^2)$ d) $m/2(R^2+r^2)$
- A uniform circular ring of radius 'R' lies in the xy plane with its centre at origin. Its moment of inertia about an axis, lying the XY-plane, parallel to X- axis and passing through a point on the Y - axis at a distance $y = 2R$ is I_1 . Its M.I. about an axis lying in a plane perpendicular to XY - plane passing through a point on the X-axis at a distance $x = d$ is I_2 . If $I_1 = I_2$, the value of 'd' is
 a) $R\sqrt{7/2}$ b) $R\sqrt{3/2}$ c) $R\sqrt{5/2}$ d) $R\sqrt{6/7}$

4. A circular plate of radius $R/2$ is cut from one edge of a thin circular plate of radius R . The moment of inertia of the remaining portion about an axis through O perpendicular to plane of the plate is (M = mass of remaining circular plate.)

- a) $\frac{13}{24}MR^2$ b) $\frac{7}{12}MR^2$
c) $\frac{13}{32}MR^2$ d) $\frac{5}{7}MR^2$

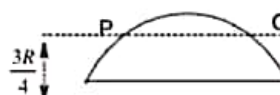


5. A uniform disc of radius 8cm lies in the x - y plane, with its centre at origin. Its moment of inertia about z -axis is equal to its moment of inertia about the line $4y = 3x + 4c$. Where c is constant. Find the value of c ($\tan 37^\circ = 0.75$)

- a) $\pm 2\text{cm}$ b) $\pm 4\text{cm}$ c) $\pm 5\text{cm}$ d) $\pm 8\text{cm}$

6. The radius of gyration of solid hemisphere of mass M and radius R about an axis parallel to the diameter at a distance $3R/4$ from center and lying in the same plane is given by (centre of mass of the hemisphere of the hemisphere lies at a height $3R/8$ from the the base)

- a) $\frac{3R}{\sqrt{10}}$ b) $\frac{5R}{4}$
c) $\frac{5R}{8}$ d) $\sqrt{\frac{2}{5}}R$



7. Moment of inertia of a uniform quarter disc of radius R and mass M about an axis through its centre of mass and perpendicular to its plane is :

- a) $\frac{MR^2}{2} - M\left(\frac{4R}{3\pi}\right)^2$ b) $\frac{MR^2}{2} - M\left(\sqrt{2}\frac{4R}{3\pi}\right)^2$ c) $\frac{MR^2}{2} + M\left(\frac{4R}{3\pi}\right)^2$ d) $\frac{MR^2}{2} + M\left(\sqrt{2}\frac{4R}{3\pi}\right)^2$

More than One correct answer Type Questions

8. A disc of mass M and radius R lies in the x - y plane with its centre at a distance a from the origin on the x -axis. Which of the following statements are correct?

- a) M.I. of the disc about x - axis is $MR^2/4$ b) M.I. of the disc about y - axis is $M\left(\frac{R^2}{2} + a^2\right)$
c) M.I. of the disc about z - axis is $M\left(\frac{R^2}{2} + a^2\right)$ d) All of the above

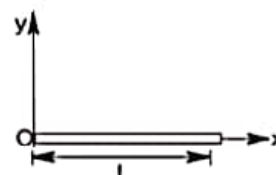
9. A rod of length L is placed along the x - axis with one of its ends at the origin O . The mass of the rod increases linearly as $\frac{dm}{dx} = \lambda x$ where λ is a constant

- a) The position of the centre of mass C of the rod, coincides with the geometrical centre C of the rod.

- b) The moment of inertia of rod about y -axis is $\frac{\lambda L^4}{4}$

- c) The moment of inertia about the perpendicular centroidal is $\frac{1}{36}\lambda L^4$ axis.

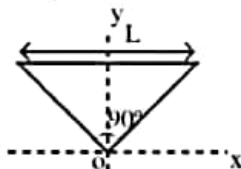
- d) All the above statemets are wrong.



Linked Comprehension Type Questions

Passage-I :

The figure shows a uniform isosceles triangular plate of mass M and base L . The angle at the apex is 90° . The apex lies at the origin and the base is parallel to x - axis.



10. The moment of inertia of the plate about the y - axis is

- a) $\frac{ML^2}{12}$ b) $\frac{ML^2}{24}$ c) $\frac{ML^2}{6}$ d) $\frac{ML^2}{4}$

11. The moment of inertia of the plate about the x - axis is

- a) $\frac{ML^2}{8}$ b) $\frac{ML^2}{32}$ c) $\frac{ML^2}{24}$ d) $\frac{ML^2}{6}$

12. The moment of inertia of the plate about its base and about the Z - axis

- a) $\frac{ML^2}{18}, \frac{ML^2}{6}$ b) $\frac{ML^2}{36}, ML^2$ c) $\frac{ML^2}{24}, \frac{ML^2}{6}$ d) $\frac{ML^2}{6}, \frac{ML^2}{24}$

Matrix Matching Type Questions

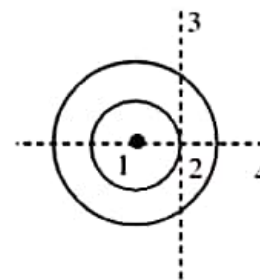
13. From a uniform disc of mass M and radius R , a concentric disc of radius $R/2$ is cut out for the remaining angular disc. I_1 is the moment of inertia about axis '1', I_2 about 2, I_3 about '3', and I_4 about '4'. Axes '1' and '2' are perpendicular to the disc and '3' and '4' are in the plane of disc. Axes '2', '3' and '4' intersect at a common point.

Column-I

- A) I_1 is equal to
B) I_2 is equal to
C) $I_3 + I_4$ is equal to
D) $I_2 - I_3$ is equal to

Column-II

- P) $\frac{21}{32}MR^2$
Q) $I_1/2$
R) $\frac{15}{32}MR^2$
S) $\frac{15}{21}MR^2$



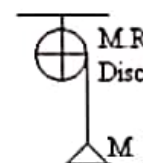
Integer Type Questions

14. Two circular loops of radii R and nR are made of same material of wire. If their M.I about their normal axis through centre are in the ratio $1 : 8$, the value of n is
15. The radius of gyration of a body about an axis at a distance of 4 cm from its centre of mass is 5 cm. The radius of gyration about a parallel axis through centre of mass is cm

16. The mass per unit area of a circular disc (of negligible thickness) varies with distance from centre 'r' as $\rho = \rho_0(1+r)$ where ρ_0 is a constant. Its moment of Inertia about an axis passing through center perpendicular to plane is $\frac{\pi\rho_0 R^4}{10}(Y+4R)$. Then $Y = \underline{\hspace{2cm}}$ [take the radius of the disc as R]

EXERCISE-II*(Torque, Toppling)***LEVEL-I (MAIN)****Straight Objective Type Questions**

- A flywheel of M.I. $6 \times 10^{-2} \text{ kgm}^2$ is rotating with an angular velocity of 20 rad s^{-1} . The torque required to bring it to rest in 4s is
 1) 1.6 Nm 2) 0.6 Nm 3) 0.8 Nm 4) 0.3 Nm
- A solid sphere and a hollow sphere of same mass and radius rotate about their diameter with same K.E. If torques of equal magnitude are applied on them, the ratio of their angular displacement before they are brought to rest is
 1) 1 : 1 2) 3 : 5 3) 5 : 3 4) 2 : 3
- For the arrangement shown in the figure, tension in the string is
 1) $\frac{Mg}{2}$ 2) $\frac{2Mg}{3}$
 3) $\frac{3Mg}{3}$ 4) $\frac{Mg}{3}$
- A uniform rod weighs 10kg and with a load 5kg attached to one end, it balances on a knife edge at 2m from that end. The length of rod is
 1) 10 m 2) 8m 3) 6m 4) 4m
- A boy and a man carry a uniform rod of length 'L' horizontally in such a way that the boy get $\frac{1}{4}$ of the load. If the boy is at one end of the rod, the distance of the man from the other end is
 1) $\frac{L}{6}$ 2) $\frac{L}{5}$ 3) $\frac{L}{4}$ 4) $\frac{L}{3}$
- A false balance is such that the beam remains horizontal when the pans are empty. An object weighs W_1 when placed in one pan and W_2 when placed in the other pan. The true weight of the object is
 1) $\frac{W_1 + W_2}{2}$ 2) $\sqrt{W_1 W_2}$ 3) $\sqrt{W_1^2 + W_2^2}$ 4) $\sqrt{\frac{W_1^2 + W_2^2}{2}}$
- A heavy wheel of radius 20 cm and weight 10kg is to be dragged over a step of height 10cm, by a horizontal force F applied at the centre of the wheel. The minimum value of F is
 1) 20 kgwt 2) 1 kgwt 3) $10\sqrt{3}$ kgwt 4) $10\sqrt{2}$ kgwt
- A particle of mass 'm' is projected with an initial velocity u at an angle ' θ ' to horizontal. The torque due to gravity on projectile at maximum height about the point of projection is
 1) $\frac{mgu^2 \sin 2\theta}{2}$ 2) $mg u^2 \sin 2\theta$ 3) $\frac{mgu^2 \sin \theta}{2}$ 4) $\frac{1}{2}mu^2 \sin 2\theta$



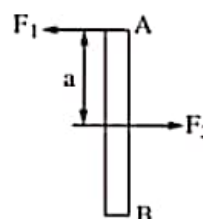
9. A cubical block of mass M and edge a slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about its centre has a magnitude

- 1) zero 2) Mga 3) $Mga \sin \theta$ 4) $\frac{1}{2}Mga \sin \theta$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

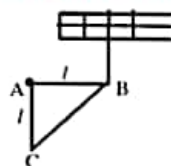
1. A thin uniform rod AB of mass 1.0 kg makes translationally motion with acceleration 2.0 m/s^2 due to two anti-parallel force F_1 and F_2 . The distance between the point at which these forces are applied is equal to $a = 20 \text{ cm}$. Besides, it is known that $F_2 = 5.0 \text{ N}$. Find the length of the rod in m.



- a) 1 m b) 2 m c) 3 m d) 1 cm
2. A rectangular block has a square base measuring $a \times a$, and its height is h . It moves with a speed v on a smooth horizontal surface
- a) It will topple if $v > \sqrt{2gh}$ b) It will topple if $v > \sqrt{2ga}$
- c) It will topple if $v > \sqrt{2ga^2/h}$ d) It will not topple for any value of v
3. A rod of length l is held vertically stationary with its lower end located at a point 'P', on the horizontal plane. When the rod is released to topple about 'P', the velocity of the upper end of the rod with which it hits the ground is

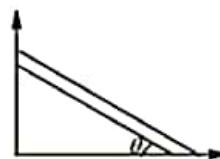
- a) $\sqrt{\frac{g}{l}}$ b) $\sqrt{3gl}$ c) $3\sqrt{\frac{g}{l}}$ d) $\sqrt{\frac{3g}{l}}$
4. A constant power is supplied to a rotating disc. The relationship of angular velocity (ω) of disc and number of rotations (n) made by the disc is governed by
- a) $\omega \propto n^{1/3}$ b) $\omega \propto n^{2/3}$ c) $\omega \propto n^{3/2}$ d) $\omega \propto n^2$
5. The door of an almirah is 6 m high, 1.5 m wide and weighs 8 kg . The door is supported by two hinges situated at a distance of 1 m from the ends. Assuming forces exerted on the hinges are equal, the magnitude of the force is $(40 + x) \text{ N}$ where ' x ' is (take $g = 9.8 \text{ m/s}^2$)
- a) 40 N b) 45 N c) 42 N d) 40 dynes
6. A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support B is

- a) $\frac{mg}{3}$ b) $\frac{2mg}{3}$
- c) $\frac{mg}{2}$ d) mg



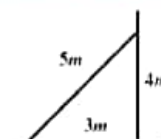
7. A uniform rod of mass M and length L is resting between a rough wall and rough floor as shown in the fig. The coefficient of friction between any two surfaces is μ . The angle θ for which the rod is just on the verge of slipping is

- a) $\tan^{-1}\left(\frac{1-\mu^2}{\mu}\right)$ b) $\tan^{-1}\left(\frac{1-\mu^2}{2\mu}\right)$
 c) $\sin^{-1}\left(\frac{1-\mu^2}{2\mu}\right)$ d) $\sin^{-1}\left(\frac{1-\mu^2}{\mu}\right)$



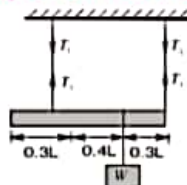
8. A uniform ladder of length 5m is placed against the wall as shown in the figure. If coefficient of friction μ is the same for both the walls, what is the minimum value of μ for it not to slip

- a) $\mu = \frac{1}{2}$ b) $\mu = \frac{1}{4}$
 c) $\mu = \frac{1}{3}$ d) $\mu = \frac{1}{5}$



9. If the bar is uniform and horizontal weighing 500N. How large must W be if T_1 and T_2 are to be equal?

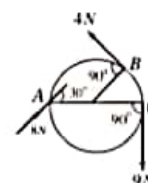
- a) 500N
 b) 300N
 c) 750N
 d) 1500N



More than One correct answer Type Questions

10. Three forces act on a wheel of radius 20cm as shown in fig.

- a) The torque produced by the force of 8N is 0.8Nm clockwise
 b) The torque produced by the force of 4N is 0.8Nm anticlockwise
 c) The torque produced by the force of 9N is zero
 d) The net torque produced by the forces is 1.8Nm clockwise



11. A ring-type flywheel of mass 100 kg and diameter 2m is rotating at the rate of $(300/\pi)$ revolutions per minute. Then:

- a) The moment of inertia of the flywheel is 100kg-m^2
 b) The kinetic energy of rotation of the flywheel is 5KJ
 c) The flywheel, if subjected to a rotating torque of 200N-m , will come to rest in 5 sec
 d) All of the above

12. A uniform cylinder of height h and radius r is placed with its circular face on a rough inclined plane and the inclination of the plane to the horizontal is gradually increased. If μ is the coefficient of friction

- a) The cylinder will slide before toppling if $\mu < \frac{2r}{h}$
 b) The cylinder will slide before toppling if μ is greater than $\frac{h}{2r}$
 c) The cylinder will topple before sliding if $\mu > \frac{2r}{h}$
 d) The cylinder will topple before sliding if μ is less than $\frac{h}{2r}$

13. The angular acceleration of the toppling pole shown in figure is given by $\alpha = k \sin \theta$, where θ is the angle between the axis of the pole and the vertical, and k is a constant. The pole starts from rest at $\theta = 0$. Choose the correct options

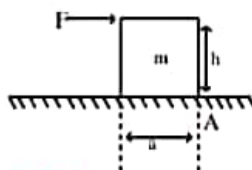
- The tangential acceleration of the upper end of the pole is $lk \sin \theta$
- The centripetal acceleration of the upper end of the pole is $2kl(1 - \cos \theta)$
- The tangential acceleration of the upper end of the pole is $2kl(1 - \cos \theta)$
- The centripetal acceleration of the upper end of the pole is $lk \sin \theta$



Linked Comprehension Type Questions

Passage-I :

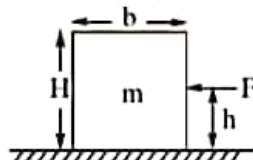
When the force F is applied the body may topple about A or it may translate.



- If the block be a cube of edge a and $\mu = 0.2$ then
 - the body will translate
 - the body will topple
 - the body may translate or topple
 - none of the above
- If the block is a cube of edge a and $\mu = 0.6$ then
 - The body will translate
 - The body will topple
 - The body first translates and then topples
 - None of the above

Passage-II :

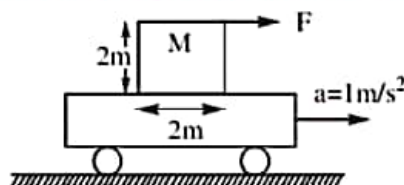
A horizontal force F is applied to a homogeneous rectangular block of mass m , width b and height H . The block moves with constant velocity and the coefficient of friction is μ_k



- What is the greatest height h at which the force F can be applied so that the block will slide without toppling?
 - $\frac{b}{\mu_k}$
 - $\frac{2b}{\mu_k}$
 - $\frac{b}{\sqrt{2}\mu_k}$
 - $\frac{b}{2\mu_k}$
- At what distance (from the left end) on the bottom face of the block will the normal reaction act when F acts at the height $H/2$? (here, assume no toppling happens)
 - $\frac{\mu_k H}{2}$
 - $\frac{b - \mu_k H}{2}$
 - $b - \frac{\mu_k H}{2}$
 - $\frac{H}{2\mu_k}$
- If the block is at rest and coefficient of static friction is μ_s , what is the height at which F has to be applied so that sliding and toppling are equally likely to occur?
 - $\frac{b}{2\mu_s}$
 - $\frac{b}{\mu_s}$
 - $\frac{b}{\sqrt{2}\mu_s}$
 - $2b\mu_s$

Matrix Matching Type Questions

19. A cubic block having square cross-section of side $2m$ and of mass $M = 10\text{kg}$ is resting over a platform moving at constant acceleration $a = 1\text{m/s}^2$. Coefficient of friction between the block and the platform is $\mu = 0.1$. A force F acts at the top of the cube as shown in Fig. Now match Column-I with Column-II

**Column-I**

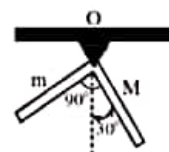
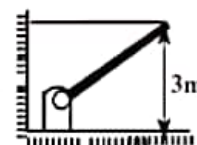
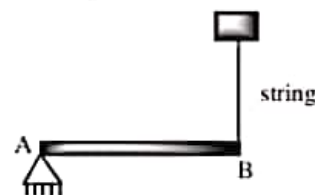
- A) $F = 0$
 B) $F = 45\text{ N}$
 C) $F = 15\text{ N}$
 D) $F = 25\text{ N}$

Column-II

- P) Block neither topples nor slips over the platform
 Q) Block topples but does not slip over the platform
 R) Block slips but does not topple over the platform
 S) Block slips as well as topples on the platform

Integer Type Questions

20. A uniform rod AB of mass m is hinged at one end A. The rod is kept in horizontal position by a massless string tied to point B. The reaction of the hinge on the end A of the rod at the instant, when string is cut is mg/x . Where ' x ' is
21. A thin uniform rod of mass m moves translationally with acceleration a due to two antiparallel force of lever arm l . One force is of magnitude F and acts at one extreme end. The length of the rod is $\frac{x(F+ma)l}{ma}$. Where ' x ' is
22. A uniform rod of mass 15kg is held stationary with the help of a light string as shown in figure. Angle made by the rod is 37° with horizontal. The tension in the string is 10^3N . Where ' x ' is
23. Two uniform rods of equal lengths but different masses are rigidly joined to form an L-shaped body, which is then pivoted as shown. If in equilibrium the body is in the shown configuration, ratio M/m will be $\sqrt{x}$. The value of x is

**EXERCISE-III**

(Angular Momentum & Law of conservation of angular momentum)

LEVEL-I (MAIN)**Straight Objective Type Questions**

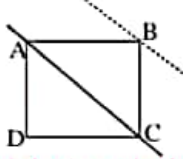
1. If the resultant of all the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that
- 1) Linear momentum of the system does not change in time
 - 2) Kinetic energy of the system does not change in time
 - 3) Angular momentum of the system does not change in time
 - 4) Potential energy of the system does not change in time

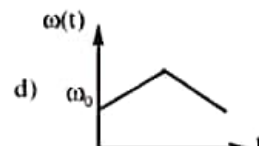
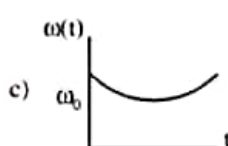
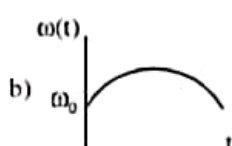
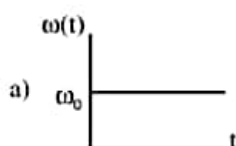
2. A Particle of mass M is moving in a straight line with uniform speed v parallel to x -axis in X - Y plane at a height ' h ' from the X -axis. Its angular momentum about the origin is
 - 1) Zero
 - 2) Mvh and is directed along positive z -axis
 - 3) Mvh and is directed along negative z -axis
 - 4) Mvh and is directed along positive x -axis
3. $\vec{F} = a\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of ' a ' for which the angular momentum is conserved is
 - 1) -1
 - 2) 0
 - 3) 1
 - 4) 2
4. A circular disc is rotating about its natural axis with angular velocity of 10 rads^{-1} . A second disc of same mass is joined to it coaxially. If the radius of disc is half of the radius of the first, then they together rotate with an angular velocity of
 - 1) 2.5 rads^{-1}
 - 2) 5 rads^{-1}
 - 3) 8 rads^{-1}
 - 4) 6 rads^{-1}
5. A man turns on rotating table with an angular speed ω . He is holding two equal masses at arms length. Without moving his arms, he just drops the two masses. How will his angular speed change?
 - 1) less than ω
 - 2) more than ω
 - 3) it will be equal to ω
 - 4) it will be more than ω if the dropped mass is more than 9.8kg and it will be less than ω if the mass dropped is less than 9.8kg
6. A thin disc rotates about an axis passing through its centre and perpendicular to its plane with a constant angular velocity ' ω '. I is the moment of inertia of that disc and ' L ' is its angular momentum about the given axis. Then rotational kinetic energy of the disc ' E ' is
 - A) $E \propto L^2$
 - B) $E \propto L^{-2}$
 - C) $E \propto I$
 - D) $E \propto I^{-1}$ (if L constant)
 - 1) A and C are correct
 - 2) B and C are correct
 - 3) B and D are correct
 - 4) A and D are correct
7. If the mass of earth and radius suddenly become 2 times and $1/4^{\text{th}}$ of the present value, the length of the day becomes
 - 1) $24h$
 - 2) $6h$
 - 3) $3/2h$
 - 4) $3h$
8. A square plate of moment of inertia ' I ' is pivoted at its centre. It has a small projection of negligible mass to one of its sides. A particle of mass ' m ' moving with a velocity v parallel to the side having projection strikes to projection and sticks to it. If ' l ' is the side length, the angular velocity of plate is
 - 1) $\frac{mv l}{2 \left[I + \frac{m l^2}{4} \right]}$
 - 2) $\frac{mv l}{2 \left[I + m l^2 \right]}$
 - 3) $\frac{mv l}{2I}$
 - 4) $\frac{mv l}{4I}$

LEVEL-II (ADVANCED)

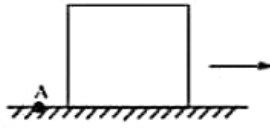
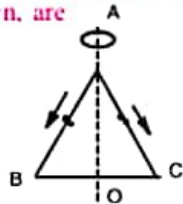
Straight Objective Type Questions

1. If the velocity of a rotating body of mass 50 kg is given by $\vec{v} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and radius vector $\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, then its angular momentum is.
 - a) $50\hat{i} + 50\hat{j} + 50\hat{k}$
 - b) $50\hat{i} + 100\hat{j} + 50\hat{k}$
 - c) $50\hat{i} + 50\hat{j} + 100\hat{k}$
 - d) $-50\hat{i} + 100\hat{j} - 50\hat{k}$
2. A shell of mass of mass ' m ' is projected with a velocity ' v ' at an angle 60° to horizontal. When it reaches the maximum height, its angular momentum with respect to point of projection is
 - a) $\frac{\sqrt{3}mv^3}{8g}$
 - b) $\frac{3mv^3}{8g}$
 - c) $\frac{3mv^3}{16g}$
 - d) $\frac{\sqrt{3}mv^3}{16g}$

3. A particle is projected with some velocity at an acute angle to the horizontal. The graph between its angular momentum (L) about the point of projection and time (t) during the flight is a/an
- a) Straight line parallel to time axis b) Parabola
c) Straight line inclined to time axis d) Ellipse
4. A small meteorite of mass m travelling towards the centre of earth strikes the earth at the equator. The earth is a uniform sphere of mass M and radius R . The length of the day was T before the meteorite struck. After the meteorite strikes the earth, the length of day increases (in sec) by
- a) $\frac{5mT}{2M}$ b) $\frac{m}{M}T$ c) $\frac{4mT}{5M}$ d) $\frac{M}{3mT}$
5. A square plate is rotating with angular velocity ω about its diagonal. Suddenly the rotation axis is shifted parallel to itself passing through B. The new angular velocity is
- a) ω b) $\frac{\omega}{2}$
c) 7ω d) $\frac{\omega}{7}$
- 
6. A thin straight tube of uniform bore is movable about its centre on a smooth horizontal table and it contains a uniform thin rod of the same length and mass whose centre of gravity is nearly at the middle point of the tube. If the system be set in motion with angular velocity ω , the angular velocity of the tube as the rod leaves it is ω/b where b is
- a) 1 b) 3 c) 7 d) 5
7. A disc of mass 100g and radius 10cm has a projection on its circumference. The mass of projection is negligible. A 20g bit of putty moving tangential to the disc with a velocity of 5ms^{-1} strikes the projection and sticks to it. The angular velocity of disc is
- a) 14.29 rads^{-1} b) 17.3 rads^{-1} c) 12.4 rads^{-1} d) 9.82 rads^{-1}
8. A boy of mass 30kg is running with a velocity of 3 ms^{-1} on ground just tangentially to a merry-go-round which is at rest. It has radius $R = 2\text{m}$, a mass of 120 kg and its radius of gyration is 1m. If the boy suddenly jumps on to the merry-go-round, the angular velocity acquired by the system is
- a) 1 rad s^{-1} b) $\frac{4}{3}\text{ rads}^{-1}$ c) $\frac{3}{4}\text{ rads}^{-1}$ d) 0.5 rad s^{-1}
9. A horizontal disc rotates freely with angular velocity ω about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is gently placed on the disc. The common angular velocity with which the two rotate, will be,
- a) ω b) $\frac{\omega}{2}$ c) $\frac{\omega}{3}$ d) $\frac{\omega}{4}$
10. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of platform. Now the platform is given an angular velocity ω_0 . When the tortoise moves along the chord of the platform with a constant velocity (with respect to the platform); The angular velocity of platform $\omega(t)$ will vary with time ' t ' as

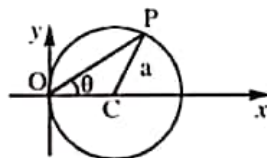


More than One correct answer Type Questions

11. A particle of mass m is rotated in the horizontal xy plane along a circular curves of radius r with angular speed ω radian per second. The rotation is opposite to that of hands of a clock
- The angular momentum of the particle about an axis passing through centre of path is $mr^2\omega\hat{k}$
 - Same as A except that $mr^2\omega\hat{k}$ accompanies a minus sign
 - Said angular momentum acts along positive z axis through the centre of circular path
 - Magnitude of said angular momentum is $\frac{1}{2}mr^2\omega$
12. A block of mass ' m ' moves on a horizontal rough surface with initial velocity ' v '. The height of the centre of mass of the block is ' h ' from the surface. Consider a point ' A ' on the surface of sliding
- Angular momentum about ' A ' is mvh initially
 - Angular momentum of the block decreases during sliding
 - Torque of the forces acting on block is zero about A
 - Angular momentum is not conserved about A
- 
13. A particle moves on a straight line with a uniform velocity. Its angular momentum (pick up the correct statement)
- is always zero
 - is zero about a point on the straight line
 - is not zero about a point away from the straight line
 - about any given point remains constant
14. A torque τ on a body about a given point is found to be equal to $\vec{C} \times \vec{L}$, where $\vec{C}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this, it follows that
- $\vec{L}$ Does not change with time
 - $\frac{d\vec{L}}{dt}$ is perpendicular to $\vec{L}$ at all instants of time
 - The magnitude of $\vec{L}$ does not change with time
 - All the above
15. An equilateral triangle ABC formed a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are not conserved as the beads slide down, are
- Angular velocity and total energy (kinetic energy and potential energy)
 - Total angular momentum and total energy
 - Angular velocity and moment of inertia about the axis of rotation
 - Total angular momentum and moment of inertia about OA
- 
16. A man of mass 100 kg stands at the rim of a turn table of radius 2 m and moment of inertia 4000 kg m² mounted on a vertical frictionless shaft at its centre. The whole system is initially at rest. The man now walks along the outer edge of the turntable with a velocity of 1 m/s relative to the earth.
- The angular velocity of turn table is 0.05rad/s along the direction of man
 - The angular velocity of turn table is 0.05rad/s opposite the direction of man
 - The turn table rotates throug $2\pi/11$ rad when man completes one revolution.
 - The angular displacement of then table is $\pi/5$ radian

Linked Comprehension Type Questions**Passage-I :**

A particle is moving in a circular path of radius a , with a constant velocity v as shown in the figure. The center of circle is marked by 'C' the particle is at P.



17. The angular momentum from the origin O can be written as:

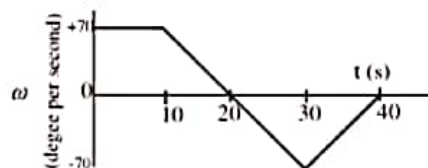
- a) $mva(1 + \cos 2\theta)$ b) $mva \cos 2\theta$ c) mva d) $mva(1 + \sin 2\theta)$

18. Angular momentum with respect to C can be written as

- a) mv b) mva c) $mva(1 + \cos 2\theta)$ d) 0

Passage-II :

Angular speed, ω is the rate at which an object rotates. Although usually measured in radians per second, ω can also be expressed in degrees per second. For instance, consider a merry-go-round. Figure shows its angular speed as a function of time t . In this graph, a positive value of ω indicates counterclockwise rotation.



A "complete revolution" occurs when the merry-go-round turns through one full circle

19. Between $t = 0$ s and $t = 10$ s, about how many revolutions does the merry-go-round complete?

- a) 1 b) 2 c) 3 d) 4

20. Between $t = 10$ s and $t = 20$ s, the merry-go-round

- a) rotates clockwise, at a constant rate
b) rotates clockwise, and slows down
c) rotates counterclockwise, at a constant rate
d) rotates counterclockwise, and slows down

21. Between $t = 30$ s and $t = 40$ s, the merry-go-round

- a) rotates clockwise, at a constant rate
b) rotates clockwise, and slows down
c) rotates counterclockwise, at a constant rate
d) rotates counterclockwise, and slows down

Matrix Matching Type Questions

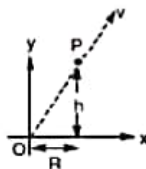
22. Column-I

- A) The change in angular momentum of a projectile relative to another projectile is
- B) The magnitude of angular momentum of a particle P of mass m about a fixed point O is

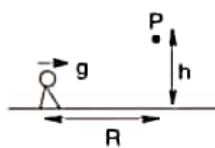
Column-II

- P) mgR
- Q) mgh

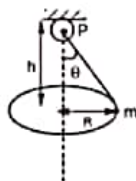
- C) $\left| \frac{d\vec{L}}{dt} \right|$ of a stationary particle P relative to a person moving with an acceleration g is



- D) $\left| \left(\frac{d\vec{L}}{dt} \right)_P \right|$ for the bob of a conical pendulum is



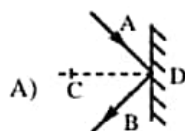
- R) Zero
- S) mvh



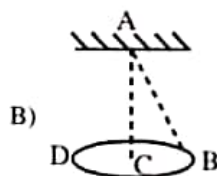
23. In the following figures identify the point w.r.t which angular momentum is constant

Column -I

Column -II



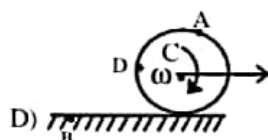
- P) A



- Q) B



- R) C



- S) D

24. A particle moves with position vector given by $\vec{r} = 3t^2\hat{i} + 4\hat{j}$ where $\vec{r}$ is measured in metres and t in seconds. For $t > 0$. Match the entries in Column-I with those in Column-II.

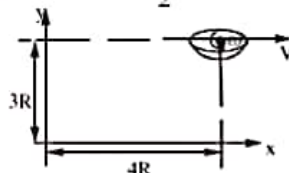
Column -I

Column -II

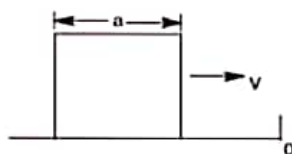
- | | |
|---|---------------------------------------|
| A) Angular acceleration (about the axis passing through the origin) | P) increases with time |
| B) Linear momentum of the particle | Q) decreases with time |
| C) Angular velocity of the particle about the axis passing through the origin | R) constant in time |
| D) Magnitude of angular momentum of the particle with respect the origin is | S) directly proportional to the time. |
| | T) is along Z axis |

Integer Type Questions

25. A disc of mass m and radius R moves in the x - y plane as shown. If the angular momentum of the disc about the origin O at the instant shown is $\frac{x}{2}mR^2\omega$ then find x . $\left[\omega = \frac{V}{R}\right]$

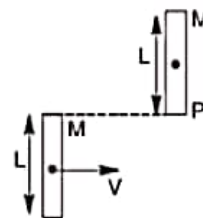
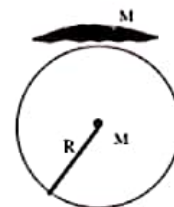
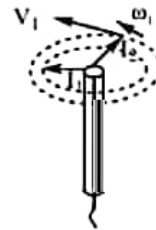


26. A boy is seated in a revolving chair revolving at an angular speed of 120 revolutions per minute. Two heavy balls from part of the revolving system and the boy can pull the balls closer to himself or may push them apart. If by pulling the balls closer, the boy decreases the moment of inertia of the system from 6 kg-m^2 to 2 kg m^2 . The new angular speed is _ revolutions per second
27. Two small balls A and B , each of mass m are joined rigidly by a light horizontal rod of length L . The rod is clamped at the centre in such a way that it can rotate freely about a vertical axis through its centre. The system is rotated with an angular speed ω about the axis. A particle P of mass m kept at rest sticks to the ball A as the ball collides with it. The new angular speed of the rod is $\frac{y\omega}{3}$ then the value of y .
28. A cubical block of side 4 cm is moving with velocity 4 m/s on a smooth horizontal plane as shown. It hits a ridge at point O . The angular speed of block after it hits O is $15Y \text{ rads}^{-1}$, then value of Y is



29. A wheel of moment of inertia 0.10 kg-m^2 is rotating about a shaft at an angular speed of 160 rev/minute . A second wheel is set into rotation at 300 re/minute and is coupled to the same shaft so that both the wheels finally rotate with a common angular speed of 200 rev/minute . The moment of inertia of the second wheel is $\frac{x}{100} \text{ kgm}^2$, then x is ____.

30. A point mass is tied to one end of a cord whose other end passes through a vertical hollow tube, caught in one hand. The point mass is being rotated in a horizontal circle of radius 2m with speed of 4 m/s . The cord is then pulled down so that the radius of the circle reduces to 1m . The ratio of kinetic energies under the final and initial state is
31. A uniform disc of mass M and radius R is supported vertically by a pivot at its centre as shown. A small dense object of mass M is attached to the rim and raised to the highest point above the centre. The unstable system is then released. The angular speed of the system when the attached object passes directly beneath the pivot is $\sqrt{\frac{xg}{3R}}$ where 'x' is
32. A bar of mass M and length L is in pure translatory motion with its centre of mass velocity V . It collides with and sticks to a second identical bar which is initially at rest. (Assume that it becomes one composite bar of length $2L$). The angular velocity of the composite bar will be $3V/xL$ counter clockwise where 'x' is



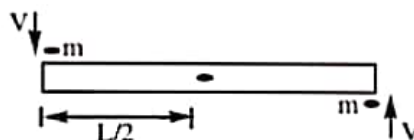
EXERCISE-IV

(Angular Impulse Rod-Particle Collisions)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A rod of mass 'm' and length l is placed on a smooth table. An another particle of same mass 'm' strikes the rod with velocity v_0 in a distance perpendicular to the rod at distance $x(<l/2)$ from its centre. Particle sticks to the rod. Let ω be the angular speed of system after collision, then the maximum possible value of impulses (by varying x) that can be imparted to the particle during collision. Particle still sticks to the rod.
- 1) $\frac{mv_0}{2}$ 2) $\frac{2mv_0}{3}$ 3) $\frac{3mv_0}{4}$ 4) $\frac{4mv_0}{5}$
2. A rigid uniform rod of mass M and length 'L' is resting on a smooth horizontal table. Two marbles each of mass 'm' and travelling with uniform speed 'V' collide with two ends of the rod simultaneously and inelastically as shown. The marbles get struck to the rod after the collision and continue to move with the rod. If $m = M/6$ and $V = L$ mts/sec, then the time taken by the rod to rotate through $\pi/2$ is



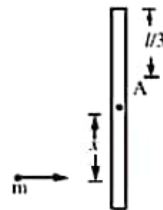
- 1) 1 s 2) $2\pi\text{ s}$ 3) $\pi\text{ s}$ 4) $\frac{\pi}{2}\text{ s}$

3. A rod of mass 'm' and length l is placed on a smooth table. An another particle of same mass 'm' strikes the rod with velocity v_0 in a distance perpendicular to the rod at distance $x(<l/2)$ from its centre. Particle sticks to the rod. Let ω be the angular speed of system after collision, then As x is increased from 0 to $l/2$, the angular speed ω :

- 1) Will continuously increase
2) Will continuously decrease
3) Will first increase and then decrease
4) Will first decrease and then increase

4. A thin uniform rod of mass 'm' and length 'l' is kept on a smooth horizontal surface such that it can move freely. Where should a particle of mass 'm' moving horizontally strike on the rod from the centre (x) such that the point 'A' at a distance $l/3$ from the end of the rod is instantaneously at rest just after the elastic collision ?

- 1) $l/2$
2) $l/3$
3) $l/6$
4) $l/4$

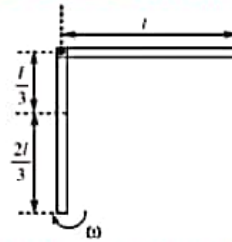


LEVEL-II (ADVANCED)

Straight Objective Type Questions

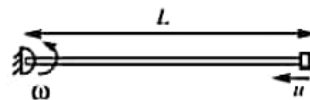
1. A thin uniform rod of mass M and length L is hinged at an end and released from rest in the horizontal position. The tension at a point located at a distance $L/3$ from the hinge point, when the rod becomes vertical, will be

- a) $\frac{22Mg}{27}$
b) $\frac{11Mg}{13}$
c) $\frac{6Mg}{11}$
d) $2Mg$



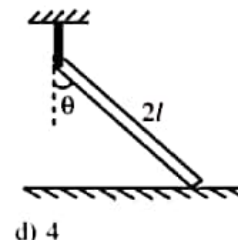
2. A small bead of mass M is launched from the end of a horizontal rod at a speed u along the rod having same mass M. The rod is free to rotate in a vertical plane about a end. Simultaneously the rod is given an angular velocity ω as shown in the figure. Find the angular acceleration of the rod. [The rod is hinged at an end]

- a) $\frac{9g}{8L}$
b) $\frac{9g}{2L} + u\omega$
c) $\frac{3}{4L} \left(\frac{3g}{2} + u\omega \right)$
d) $\frac{3}{4L} \left(\frac{3g}{2} - 2u\omega \right)$



3. A uniform rod of length $2l$ and mass 'm' is suspended from one end by inextensible string and other end lies on smooth ground. The angle made by rod with vertical is $\theta = \sin^{-1}(1/\sqrt{3})$. If N_1 and N_2 represents the contact force from ground on rod just before and just after cutting string then find the ratio of N_1/N_2

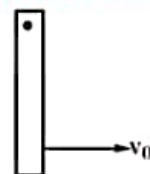
- a) 1
b) $1/2$
c) $3/2$



8. A unknown solid sphere of mass m and radius r is kept on rough horizontal surface with coefficient of friction $\mu = \frac{1}{3}$ as shown in the figure. An impulse $J = MV_0$ is applied tangentially on the sphere at an angle 37° with the horizontal as shown in the figure. (Assuming $J \gg Mgd$, $g = 10 \text{ m/s}^2$)



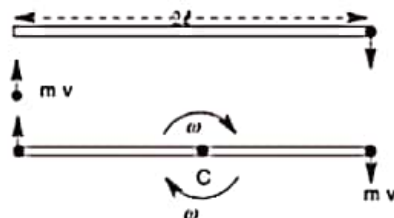
- a) The velocity of centre of mass of the sphere just after impulse applied is V_0
 b) The angular velocity of sphere just after impulse applied is $\frac{2V_0}{R}$
 c) The velocity of centre of mass of the sphere just after impulse applied is $\frac{4V_0}{5}$
 d) The angular velocity of sphere just after impulse applied is $\frac{V_0}{R}$
9. A thin uniform rigid rod of length ' l ' is hinged at one end so that it can move in a vertical plane by rotating about a horizontal axis through upper end. The lower end is given a sharp blow and made to acquire a linear velocity v_0 . Maximum height attained by lower end of the rod from the initial position is :
- a) $2l$ for $v_0 \geq \sqrt{4gl}$ b) $2l$ for $v_0 \geq \sqrt{5gl}$
 c) $2l$ for $v_0 \geq \sqrt{6gl}$ d) l for $v_0 = \sqrt{3gl}$



Linked Comprehension Type Questions

Passage-I :

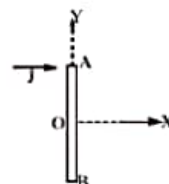
Two skaters, each of mass 50kg , approach each other along parallel paths separated by 3m . They have equal and opposite velocities of 10ms^{-1} . The first skater carries a long light pole 3m long and the second skater grabs the end of it as he passes on the frictionless surface and the two skaters are connected by the pole.



10. The angular velocity of the pole after the skaters are connected is :
- a) $\frac{22}{3}\text{rads}^{-1}$ b) $\frac{20}{3}\text{rads}^{-1}$ c) $\frac{17}{3}\text{rads}^{-1}$ d) $\frac{14}{3}\text{rads}^{-1}$
11. By pulling on the pole, the skaters reduce their distance apart to 1m . The angular velocity now is and the ratio of K.E. of system in above two cases is $\frac{KE_f}{KE_i} =$
- a) 45 rad s^{-1} , 7 b) 50 rad s^{-1} , 10 c) 55 rad s^{-1} , 9 d) 60 rad s^{-1} , 9

Passage-II :

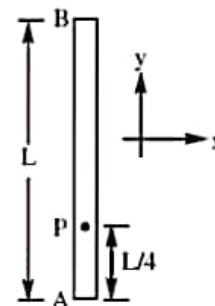
A rod AB of length 2m and mass 2kg is lying on a smooth horizontal x-y plane with its centre at origin O as shown in figure. An impulse J of magnitude 10Ns is applied perpendicular to AB at end A.



12. The distance of point P from the centre of the rod which is at rest just after the impact is
 a) $1/4$ m b) $1/2$ m c) $1/3$ m d) $2/3$ m
13. Find the coordinates of point A of the rod after time $t = \pi/45$ of the impact if the centre of mass of the rod is taken as origin at that time.
 a) $\left(\frac{\sqrt{3}}{2}m, \frac{1}{2}m\right)$ b) $\left(\frac{\sqrt{3}}{2}m, 1m\right)$ c) $\left(\frac{3}{4}m, \frac{3}{2}m\right)$ d) None of these
14. If the rod is hinged at B and instead of impulse J, a point masses of 1kg moving with a speed 10m/s collides at A and sticks to it, then find the angular velocity of the rod just after the collision
 a) 4 rad/s b) 3 rad/s c) 2 rad/s d) 1 rad/s

Passage-III :

Uniform rod AB hinged about a fixed point 'P' is initially vertical. Rod is released from vertical position. When rod is in horizontal position.



15. The acceleration of centre of mass of rod is
 a) $-\frac{6g}{7}\hat{i} - \frac{3g}{7}\hat{j}$ b) $-\frac{12g}{7}\hat{i} - \frac{6g}{7}\hat{j}$
 c) $-\frac{3g}{7}\hat{i} - \frac{9g}{7}\hat{j}$ d) $-\frac{9g}{7}\hat{i} - \frac{3g}{7}\hat{j}$
16. The acceleration of end B of rod is
 a) $-\frac{6g}{7}\hat{i} - \frac{12g}{7}\hat{j}$ b) $-\frac{5g}{7}\hat{i} - \frac{9g}{7}\hat{j}$ c) $-\frac{18g}{7}\hat{i} - \frac{5g}{7}\hat{j}$ d) $-\frac{18g}{7}\hat{i} - \frac{9g}{7}\hat{j}$
17. Reaction on hinge is
 a) $-\frac{8mg}{7}\hat{i} - \frac{12mg}{7}\hat{j}$ b) $-\frac{3mg}{7}\hat{i} - \frac{9mg}{7}\hat{j}$ c) $-\frac{6mg}{7}\hat{i} + \frac{4mg}{7}\hat{j}$ d) $-\frac{12mg}{7}\hat{i} - \frac{6mg}{7}\hat{j}$

Matrix Matching Type Questions

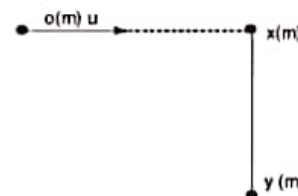
18. Two particles X and Y of mass m each, are joined by a rigid massless rod of length l . A particle O of mass m moving with a speed of u normal to X and Y strikes X and stick to it. The centre of mass of the 'X + Y + O' system is K

Column-I

- A) Velocity of K before impact
 B) Velocity of K after impact
 C) Velocity of 'X + O' immediately after impact
 D) Velocity of Y immediately after impact

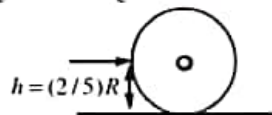
Column-II

- P) 0
 Q) $\frac{u}{2}$
 R) u
 S) $\frac{u}{3}$

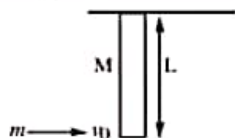


19. Column-I

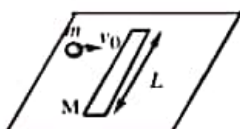
- A) An impulsive force acts at a height $h = (2/5)R$ on a solid sphere, lying on a rough surface



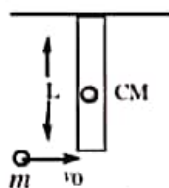
- B) A small particle of mass m moving horizontally collides with a vertical hinged rod inelastically $e = 0$



- C) A small particle of mass m strikes elastically at the end of a horizontal rod kept on a smooth surface



- D) A small particle of mass m collides with a vertical rod elastically



Column-II

- P) Momentum is conserved

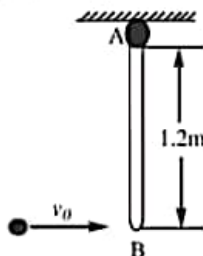
- Q) Angular momentum is conserved

- R) Energy is conserved

- S) Momentum increases

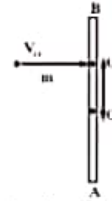
Integer Type Questions

20. A uniform rod of length L rests on a frictionless horizontal surface. The rod is pivoted about a fixed frictionless axis at one end. The rod is initially at rest. A bullet travelling parallel to the horizontal surface and perpendicular to the rod with speed V_0 strikes the rod at its centre and becomes embedded in it. The mass of the bullet is one-sixth of the mass of the rod. The final angular velocity of the rod is ω and the ratio of the kinetic energy of the system after collision to the kinetic energy of the bullet before collision is $1/x$ where x is
21. A 2 kg sphere moving horizontally to the right with an initial velocity of 5 m/s strikes the lower end of an 8 kg rigid rod AB. The rod is suspended from a hinge at A and is initially at rest. Knowing that the coefficient of restitution between the rod and sphere is 0.80. The velocity of the sphere immediately after the impact is $V_s = \frac{1}{x} \text{ ms}^{-1} (\text{left})$ where 'x' is



22. A uniform rod AB of length l and mass m is lying on a smooth table.

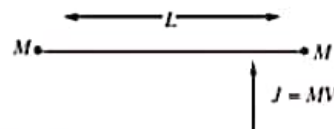
A small particle of mass m strikes the rod with a velocity V_0 at C at a distance x from the centre O. The particle comes to rest after collision. The value of x , so that point A of the rod remains stationary just after collision is l/y . The value of y is



23. A uniform rod AB of mass $3m$ and length $4l$, which is free to turn in a vertical plane about a smooth horizontal axis through A, is released from rest when horizontal. When the rod first becomes vertical, a point C of the rod, where $AC = 3l$, strikes a fixed peg. Exerted by the peg on the rod if the rod

is brought to rest by the peg is $\frac{x}{3}m\sqrt{3gl}$ where 'x' is

24. Consider a body, shown in figure, consisting of two identical balls, each of mass M connected by a light rigid rod. If an impulse $J = MV$ is imparted to the body at one of its ends, its angular velocity would be $\frac{KV}{L}$. The value of K is



EXERCISE-V

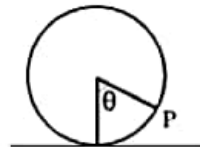
(Rolling)

LEVEL-I (MAIN)

Straight Objective Type Questions

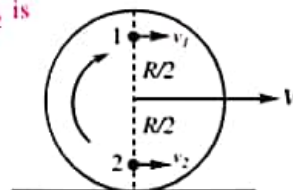
1. A disc rolls on ground without slipping. Velocity of centre of mass is v . If the point P lies on the circumference of disc at angle θ as shown in the figure then the speed v_p at the point P, when $\theta = 90^\circ, 120^\circ, 180^\circ$

- 1) $\sqrt{2}v, \sqrt{3}v, 2v$
- 2) $v, 2v, \sqrt{3}v$
- 3) $2v, \sqrt{3}v, \sqrt{2}v$
- 4) $\sqrt{3}v, v, \sqrt{5}v$



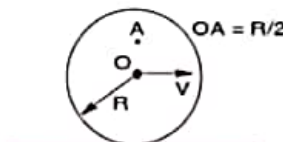
2. Disc is in pure rolling as shown below; then v_1/v_2 is

- 1) 1 : 3
- 2) 3 : 1
- 3) 2 : 3
- 4) 1 : 1

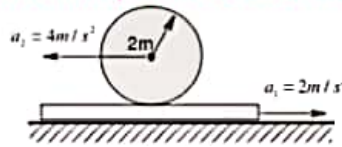


3. A sphere is performing pure rolling as shown in the figure. The radius of curvature of trajectory of point A in the shown position is XR . Find $2x$.

- 1) 9
- 2) 4
- 3) 1
- 4) 0



4. In figure, a sphere of radius $2m$ rolls on a plank. The value of α is (where α is angular acceleration)



- 1) 2 rad/s^2 2) 3 rad/s^2 3) 4 rad/s^2 4) zero rad/s^2
5. Consider a cylinder of mass M resting on a rough horizontal rug that is pulled out from under it with acceleration ' a ' perpendicular to the axis of the cylinder. What is F_{friction} at point P? It is assumed that the cylinder does not slip.
- 1) $\frac{Ma}{3}$ 2) Mg 3) $\frac{Ma}{2}$ 4) Ma
6. A uniform solid cylindrical roller of mass ' m ' is being pulled on a horizontal surface with force F parallel to the surface and applied at its centre. If the acceleration of the cylinder is ' a ' and it is rolling without slipping then the value of ' F ' is:
- 1) $2ma$ 2) $\frac{3}{2}ma$ 3) ma 4) $\frac{5}{3}ma$
7. A disc at rest at the top of an inclined plane of height ' h ' rolls down without slipping and acquires a velocity ' v ' on reaching the bottom. If the same disc slides down a smooth incline and acquires the same velocity on reaching the bottom the height of smooth incline is
- 1) $h/3$ 2) $h/2$ 3) $2h/3$ 4) h

LEVEL-II (ADVANCED)

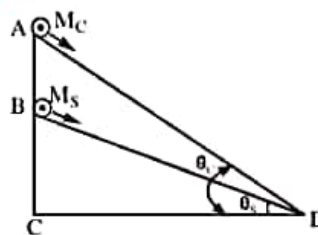
Straight Objective Type Questions

1. A sphere rolls without slipping on an incline of inclination θ . The minimum coefficient of static friction to support pure rolling is to be
- a) $\frac{2}{3}\tan\theta$ b) $\frac{2}{5}\tan\theta$ c) $\frac{2}{7}\tan\theta$ d) $\frac{3}{7}\tan\theta$
2. A wheel of radius ' r ' rolls without slipping with a speed ' v ' on a horizontal road. When it is at point 'A' on the road, a small lump of mud separates from the wheel at its highest point B and drops at point 'C' on the road. The distance AC will be
- a) $v\sqrt{\frac{r}{g}}$ b) $2v\sqrt{\frac{r}{g}}$ c) $4v\sqrt{\frac{r}{g}}$ d) $v\sqrt{\frac{3r}{g}}$
3. A disc is at rest at the top of a rough inclined plane. It rolls without slipping. At the bottom of the inclined plane there is a vertical groove of radius ' R '. In order to loop the groove, the minimum height of incline required is
- a) $\frac{15R}{4}$ b) $\frac{9R}{4}$ c) $\frac{5R}{2}$ d) $\frac{7R}{5}$
4. A solid cylinder of mass ' m ' rolls without slipping down an inclined plane making an angle θ with the horizontal. The frictional force between the cylinder and the incline is
- a) $mg \sin\theta$ b) $\frac{mg \sin\theta}{3}$ c) $mg \cos\theta$ d) $\frac{2mg \sin\theta}{3}$

5. A cylinder of mass M_c and sphere of mass M_s are placed at points A and B of two inclines, respectively. (See figure). If they roll on the incline without slipping such that their accelerations are the same,

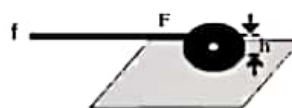
then the ratio $\frac{\sin \theta_c}{\sin \theta_s}$ is :

- a) $\frac{8}{7}$ b) $\sqrt{\frac{15}{14}}$
c) $\frac{15}{14}$ d) $\sqrt{\frac{8}{7}}$



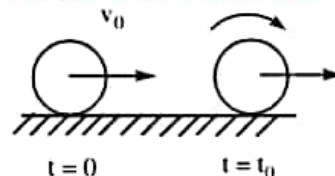
6. A billiard ball of radius R is struck by a cue as shown. Assume that the impulse given by the cue is purely horizontal. For the ball to roll without any initial slippage

- a) $h = \frac{5}{2}R$ b) $h = \frac{2}{3}R$
c) $h = \frac{2}{5}R$ d) $h = \frac{3}{2}R$



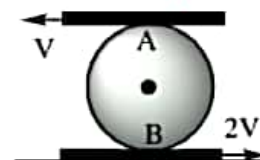
7. A uniform disc of mass m and radius R is projected horizontally with velocity v_0 on a rough horizontal floor, so that it starts off with a purely sliding motion at $t = 0$. After t_0 seconds, it acquires a purely rolling motion as shown in figure. The velocity of the centre of mass of the disc at t_0

- a) $\frac{3}{2}v_0$ b) $\frac{3}{4}v_0$
c) $\frac{2}{3}v_0$ d) $\frac{4}{3}v_0$



8. The cylinder of the radius r is confined to roll without slipping at A and B. If the plates have the velocities shown, determine the angular velocity of the cylinder.

- a) $\frac{3v}{2R}$ b) $\frac{6v}{R}$
c) $\frac{7v}{R}$ d) $\frac{9v}{R}$

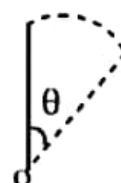


9. A string is wrapped around a wheel of radius ' r '. The axis of the wheel is horizontal and its M.I. about the axis is I . Weight ' mg ' is tied to free end of the string which is released to fall down from rest position. The angular velocity of the wheel after it has fallen through distance ' h ' will be

- a) $\left(\frac{2gh}{I + mr^2}\right)^{\frac{1}{2}}$ b) $\left(\frac{2mgh}{I + mr^2}\right)^{\frac{1}{2}}$ c) $\frac{2mgh}{I + 2mr^2}$ d) $\sqrt{2gh}$

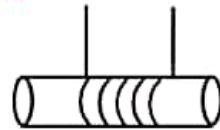
10. A vertical rod is free to rotate about 'O' as shown. It begins to rotate. When it has turned through an angle ' θ ', its angular velocity ' ω ' is given as

- a) $\sqrt{\frac{6g}{\ell}} \sin \theta$ b) $\sqrt{\frac{6g}{\ell}} \sin \left(\frac{\theta}{2}\right)$
c) $\sqrt{\frac{6g}{\ell}} \cos \frac{\theta}{2}$ d) $\sqrt{\frac{6g}{\ell}} \cos \theta$



11. A cylinder of mass M and radius R starts falling under gravity at $t = 0$ as shown in figure. If the mass of chord is negligible, the tension in each string is

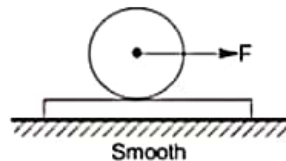
- a) $\frac{Mg}{6}$ b) $\frac{Mg}{4}$
c) $\frac{Mg}{2}$ d) Mg



More than One correct answer Type Questions

12. A disc of mass m and radius R is placed over plank of mass m . There is sufficient friction between disc & plank to prevent slipping. A force F is applied at centre of disc then.

- a) Acceleration of plank is $\frac{F}{2m}$
b) Acceleration of plank is $\frac{F}{4m}$
c) Friction force between disc and plank is $F/4$ d) Friction force between disc and plank is $F/3$

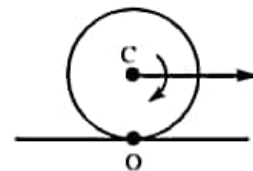


13. A uniform circular disc of radius R is rolling on a horizontal surface. If v be velocity of its CM and ω be its angular velocity about CM such that $v > R\omega$, then which of the following are correct ?

- a) No point on the disc will have velocity less than $v - R\omega$
b) No point on the disc will have velocity greater than $v + R\omega$
c) No point on the disc other than CM will have velocity equal to v
d) There are atleast two points on the disc having velocity equal to $\sqrt{v^2 + R^2\omega^2}$

14. Regarding pure rolling motion of a spherical body of radius R shown, the correct statement(s) are

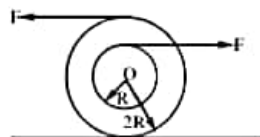
- a) When a rigid body rolls, without slipping on a horizontal surface a point on its circumference moves along a curved path called cycloid.
b) The distance covered by the point 'O' on the circumference of the rolling object along the cycloid is $8R$ in one revolution.
c) The distance moved by the center of mass in one revolution is $2\pi R$
d) The distance moved by the center of mass in one revolution is less than $2\pi R$



Linked Comprehension Type Questions

Passage-I :

In the given figure $F = 10\text{N}$, $R = 1\text{ m}$, mass of the body is 2 kg and moment of inertia of the body about an axis passing through O and perpendicular to plane of body is 4 kgm^2 . O is the centre of mass of the body.



15. If the ground is smooth what is total kinetic energy of the body after 2 s :

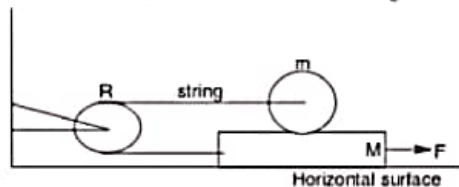
- a) 25 J b) 50 J c) 75 J d) 100 J

16. If ground is sufficiently rough, what is kinetic energy of the body now in the given time interval.

- a) 10.33 J b) 25.67 J c) 16.67 J d) None of these

Passage-II :

A solid cylinder of mass m and radius R and that of plank M placed on smooth horizontal surface. Given that sufficient friction is present between cylinder and the plank surface to prevent sliding of cylinder. If plank is pulled with force F , answer the following



17. The acceleration of the cylinder is

- a) $\frac{F}{M+3m}$ b) $\frac{2F}{M+3m}$ c) $\frac{F}{3M+m}$ d) $\frac{2mF}{3M+m}$

18. The tension in the string is

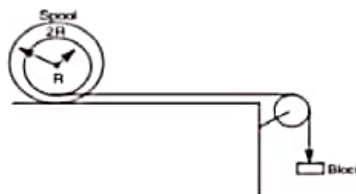
- a) $\frac{mF}{M+3m}$ b) $\frac{2mF}{M+3m}$ c) $\frac{mF}{3M+m}$ d) $\frac{2mF}{3M+m}$

19. The magnitude of friction between cylinder and plank is

- a) $\frac{2mF}{M+3m}$ b) $\frac{mF}{3M+m}$ c) $\frac{mF}{M+3m}$ d) $\frac{2mF}{3M+m}$

Passage-III :

In the diagram a block of mass m is connected to a spool of mass $3m$, inner radius R and outer radius $2R$ as shown. If pulley is light and surface is sufficiently rough such that no sliding takes place. Let moment of inertia of spool is mR^2 , about the axis passing through centre of mass and perpendicular to plane of rotation. The system is released from rest, answer the following



20. The acceleration of the block

- a) $\frac{9g}{14}$ b) $\frac{5g}{14}$ c) $\frac{3g}{14}$ d) $\frac{g}{14}$

21. The angular acceleration of spool is

- a) $\frac{g}{14R}$ b) $\frac{3g}{14R}$ c) $\frac{5g}{14R}$ d) $\frac{9g}{14R}$

22. When the block falls down by a distance x the angular speed of spool is

- a) $\sqrt{\frac{gx}{3R^2}}$ b) $\sqrt{\frac{gx}{5R^2}}$ c) $\sqrt{\frac{gx}{7R^2}}$ d) $\sqrt{\frac{gx}{9R^2}}$

Matrix Matching Type Questions

23. Column-I

Column-II

- A) 
A spinning ball lowered on a rough surface

- B) 
A ball is projected with velocity v_0 on a rough surface

- C) 
A ball projected with velocity $v_0 = R\omega_0$ projected on a smooth horizontal surface

- D) 
A spool pulled by a force F , rolling without slipping on a rough surface

P) Net workdone by friction force is negative

Q) Rotational workdone by force is negative

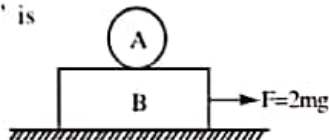
R) Translational workdone friction is positive

S) Net workdone by friction is zero.

Integer Type Questions

24. A solid sphere of mass 1.5 kg rolls on a frictionless horizontal surface, its centre of mass moving at a speed of 5ms^{-1} . Then it rolls up an incline of 30° to horizontal. If the height attained by the sphere before it stops is $(g=10\text{ms}^{-2})\left(\frac{7}{x}\right)\text{m}$ then find x .

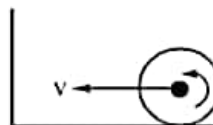
25. In the figure shown a block B of mass m can slide on a fixed horizontal smooth plane. A solid sphere A of radius r of the same mass rolls without sliding on the block B. The angular acceleration of the sphere is $10g/xr$ where 'x' is



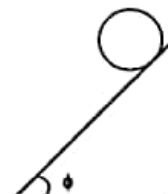
26. A system of uniform cylinders and plates is shown. All the cylinders are identical and there is no slipping at any contact. Velocity of lower and upper plates is v and $2v$ respectively as shown. Then the ratio of angular speeds of the upper cylinders to lower cylinders is



27. A uniform circular disc of mass m and radius R is rolling without slipping on a smooth horizontal surface as shown in figure. The disc collides with a rough vertical wall having coefficient of restitution 0.6. If the disc rolls without slipping after collision with the wall, and the coefficient of friction is $n/10$, then the value of 'n' is



28. A hollow sphere of mass 100g is released from the top of an inclined plane of inclination $\phi (=45^\circ)$ and height $h (=8\text{ m})$. The coefficient of friction between the plane and the sphere is 0.25. The work done (in J) by the force of friction as the sphere rolls down the inclined plane is



KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

- 1) 1 2) 1 3) 3 4) 1 5) 1

LEVEL - II

- 1) a 2) d 3) a 4) a 5) c 6) d 7) b 8) ac
-
- 9) bc 10) b 11) c 12) c 13) A-R; B-P; C-P; D-Q 14) 2
-
- 15) 3 16) 5

EXERCISE-II

LEVEL - I

- 1) 4 2) 1 3) 4 4) 3 5) 4 6) 2 7) 3 8) 4
-
- 9) 4

LEVEL - II

- 1) a 2) d 3) b 4) a 5) c 6) b 7) b 8) c
-
- 9) d 10) abd 11) abcd 12) ac 13) ab 14) a 15) b 16) d
-
- 17) b 18) a 19) A-P; B-R; C-P, D-R 20) 4 21) 2 22) 2
-
- 23) 3

EXERCISE-III

LEVEL - I

- 1) 1 2) 3 3) 1 4) 3 5) 3 6) 4 7) 4 8) 1

LEVEL - II

- 1) d 2) c 3) b 4) a 5) d 6) d 7) a 8) c
-
- 9) c 10) b 11) ac 12) abd 13) bcd 14) bc 15) acd 16) bcd
-
- 17) a 18) b 19) b 20) d 21) b 22) A-R; B-R; C-Q; D-P
-
- 23) A-RS; B-R; C-P; D-Q 24) A-T; B-PS; C-T; D-PS 25) 5
-
- 26) 6 27) 2 28) 5 29) 4 30) 4 31) 8 32) 4

EXERCISE-IV

LEVEL - I

- 1) 1 2) 4 3) 3 4) 1

LEVEL - II

- 1) d 2) d 3) a 4) abc 5) abc 6) abc 7) abc 8) ab
-
- 9) cd 10) b 11) d 12) c 13) a 14) b 15) a 16) d
-
- 17) c 18) A-S; B-S; C-Q; D-P 19) A-S; B-P; C-R; D-Q 20) 9
-
- 21) 7 22) 6 23) 8 24) 1

EXERCISE-V

LEVEL - I

- 1) 1 2) 2 3) 1 4) 2 5) 1 6) 2 7) 3

LEVEL - II

- 1) c 2) c 3) a 4) b 5) c 6) c 7) c 8) a
-
- 9) b 10) b 11) a 12) bc 13) ab 14) bc 15) b 16) c
-
- 17) a 18) b 19) c 20) b 21) a 22) c
-
- 23) A-PQR; B-P; C-S; D-S 24) 4 25) 9 26) 3 27) 5 28) 1